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Construction machine and operation method

Abstract

This disclosure relates to a construction machine utilizing battery and hydraulic power and comprising an electronic power unit comprising a battery and an electric motor, a hydraulic pump driven by the electronic power unit and discharging hydraulic fluid, a hydraulic line through which the hydraulic fluid discharged from the hydraulic pump moves, a main control valve installed on the hydraulic line and controlling the supply of the hydraulic fluid to at least one of propulsion devices or various working devices requiring hydraulic power, a bypass cut valve installed downstream of the main control valve on the hydraulic line and configured to open and close the hydraulic line, and a controller initiating a warm-up mode to increase the output of the electronic power unit and open the bypass cut valve.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is the National Stage filing under 35 U.S.C. 371 of International Application No. PCT/KR2022/008924, filed on Jun. 23, 2022, which claims the benefit of earlier filing date of and right of priority to Korean Application No. 10-2021-0082572 filed on Jun. 24, 2021, the contents of which are all hereby incorporated by reference herein in their entirety.

FIELD

(2) This disclosure relates to a construction machine utilizing battery and hydraulic power and operation method thereof.

BACKGROUND

(3) Construction machines refer to all machines used in civil engineering or building construction projects. Typically, construction machines are equipped with engines, such as diesel engines, to obtain hydraulic power for both propulsion and operational devices, harnessing the engine's power to drive a hydraulic pump and enabling the machine's movement or operation through the force generated by the hydraulic pump.

(4) These construction machines, equipped with diesel engines, have been plagued by high operating costs, attributed not only to the expensive fuel costs caused by the depletion of fossil fuels, particularly diesel, and the pressing need for alternative energy sources but also to the high fuel consumption inherent in the nature of heavy equipment.

(5) To address these drawbacks, research is actively being conducted on construction machines powered by batteries. Construction machines equipped with batteries operate by driving an electric motor using electrical energy generated from the batteries, instead of an engine, to power the hydraulic pump.

SUMMARY

(6) Construction machines relying on battery and hydraulic power face the drawback of experiencing reduced performance in low-temperature conditions, such as winter, when the temperature drops below 0° C.

(7) Specifically, in low-temperature conditions, the output of the battery, which serves as the energy source, is reduced. Moreover, in low-temperature conditions, the viscosity of the hydraulic fluid increases, leading to decreased fluidity, which results in a reduction of the output power generated through the hydraulic pump.

(8) The objective of this disclosure is to prevent the reduction in performance of a construction machine utilizing battery and hydraulic power in low-temperature conditions.

(9) To solve the problems described above, the present disclosure has embodiments with the following features.

(10) According to an embodiment, a construction machine includes an electronic power unit comprising a battery and an electric motor, a hydraulic pump driven by the electronic power unit and discharging hydraulic fluid, a hydraulic line through which the hydraulic fluid discharged from the hydraulic pump moves, a main control valve installed on the hydraulic line and controlling the supply of the hydraulic fluid to at least one of propulsion devices or various working devices requiring hydraulic power, a bypass cut valve installed downstream of the main control valve on the hydraulic line and configured to open and close the hydraulic line, and a controller initiating a warm-up mode to increase output of the electronic power unit and open the bypass cut valve.

(11) The controller initiates the warm-up mode based on temperature of the battery being lower than a first threshold value.

- (12) The controller, during the warm-up mode, controls the battery by changing discharge rate of the battery based on the temperature of the battery.
- (13) The controller initiates the warm-up mode based on temperature of the hydraulic fluid being lower than or equal to a second threshold value.
- (14) The controller, during the warm-up mode, additionally activates a heating device based on the temperature of the hydraulic fluid being lower than or equal to the second threshold value.
- (15) The controller initiates the warm-up mode based on a pilot cut-off signal being input from a pilot cut-off lever.
- (16) The controller controls the construction machine to generate an alarm based on the pilot cut-off signal being not input.
- (17) According to an embodiment, a method for controlling a construction machine including an electronic power unit including a battery and an electric motor, a hydraulic pump driven by the electronic power unit and discharging hydraulic fluid, a hydraulic line through which the hydraulic fluid discharged from the hydraulic pump moves, a main control valve installed on the hydraulic line and controlling the supply of the hydraulic fluid to at least one of propulsion devices or various working devices requiring hydraulic power, a bypass cut valve installed downstream of the main control valve on the hydraulic line and configured to open and close the hydraulic line, and an oil tank storing the hydraulic fluid to be supplied to the hydraulic pump and recovering the hydraulic fluid discharged by the hydraulic pump and moved along the hydraulic line, includes initiating a warm-up mode to increase output of the electronic power unit and open the bypass cut valve.
- (18) The initiating of the warm-up mode includes initiating the warm-up mode based on temperature of the battery being lower than a first threshold value.
- (19) The initiating of the warm-up mode includes controlling the battery by changing discharge rate of the battery based on the temperature conditions of the battery.
- (20) The initiating of the warm-up mode includes initiating the warm-up mode based on temperature of the hydraulic fluid being lower than or equal to a second threshold value.
- (21) The initiating of the warm-up mode includes activating a heating device additionally based on the temperature of the hydraulic fluid being lower than or equal to the second threshold value.
- (22) The initiating of the warm-up mode includes initiating the warm-up mode based on a pilot cut-off signal being input from the pilot cut-off lever.
- (23) The initiating of the warm-up mode includes controlling the construction machine to generate an alarm based on the pilot cut-off signal being not input.
- (24) The present disclosure is advantageous in terms of prevent the reduction in performance of a construction machine utilizing battery and hydraulic power in low-temperature conditions.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a configuration diagram of a construction machine according to an embodiment;
(2) FIG. 2 is a block diagram of an electronic power unit according to an embodiment; and
(3) FIG. 3 is a flowchart illustrating a control method of a controller.

DETAILED DESCRIPTION

(4) Hereinafter, embodiments of the present disclosure will be described with reference to accompanying drawings. In the specification, when a component (or area, layer, part, etc.) is mentioned as being “on top of,” “connected to,” or “coupled to” another component, it means that it may be directly connected/coupled to the other component, or a third component may be placed between them.

(5) The same reference numerals refer to the same components. In addition, in the drawings, the thickness, proportions, and dimensions of the components are exaggerated for effective description

of the technical content. The expression “and/or” is taken to include one or more combinations that can be defined by associated components.

(6) The terms “first,” “second,” etc. are used to describe various components, but the components should not be limited by these terms. The terms are used only for distinguishing one component from another component. For example, a first component may be referred to as a second component and, similarly, the second element may be referred to as the first component, without departing from the scope of the present disclosure. As used herein, the singular forms are intended to include the plural forms as well, unless the context clearly indicates otherwise.

(7) The terms such as “below,” “lower,” “above,” “upper,” etc. are used to describe the relationship of components depicted in the drawings. The terms are relative concepts and are described based on the direction indicated on the drawing.

(8) It will be further understood that the terms “comprises,” “has,” and the like are intended to specify the presence of stated features, numbers, steps, operations, components, parts, or a combination thereof but are not intended to preclude the presence or possibility of one or more other features, numbers, steps, operations, components, parts, or combinations thereof.

(9) Hereinafter, a description is made of the construction machine according to an embodiment of the present disclosure with reference to FIG. 1. In an embodiment of the present disclosure, the explanation is made with an excavator as an example of the construction machine. However, the embodiments of the present disclosure are not limited to this example and can be applied to any construction machine that transfers power using hydraulic fluid discharged by a hydraulic pump.

(10) As illustrated in FIG. 1, the construction machine according to an embodiment of the present disclosure includes a hydraulic pump **800**, an electronic power unit **200**, a hydraulic line **610**, an oil tank **850**, a main control valve (MCV) **500**, a bypass cut valve **400**, a pilot cut-off lever **300**, and a controller **700**.

(11) The construction machine according to an embodiment of the present disclosure may further include various working devices and propulsion devices. The construction machine may also include operating devices such as joysticks, levers, and pedals installed in the operator's cab to allow the operator to operate the various working devices **170** and propulsion devices **160**. The pilot cut-off lever **300** may also be one of operating devices within the operating apparatus.

(12) The electronic power unit **200** includes a battery and an electric motor to generate power. The electronic power unit **200** supplies rotational power to the hydraulic pump **800**.

(13) The hydraulic pump **800** operates with the power generated by the electronic power unit **200** and discharge hydraulic fluid. The hydraulic fluid discharged from the hydraulic pump **800** is supplied to the propulsion devices **160** including a propulsion motor for travel, a swing motor **180** used for the swing of the upper rotating body, and driving devices including boom cylinders, arm cylinders, bucket cylinders, and optional cylinders used for the various working devices **170**. The driving devices are powered by the hydraulic fluid supplied from the hydraulic pump **800**.

(14) In an embodiment of the present disclosure, the hydraulic pump **800** may be equipped with an angle sensor (not shown) capable of measuring the swashplate angle and electrically controlled by an electrical signal generated by the controller **700**. Here, the information measured by the angle sensor is transmitted to the controller **700**. Accordingly, the controller **700** may forcibly adjust the swashplate angle of the hydraulic pump **800** based on the information from the angle sensor. That is, the hydraulic pump **800** may be forcibly operated solely by the electrical signal generated by the controller **700**.

(15) The hydraulic line **610** transports the hydraulic fluid discharged from the hydraulic pump **800** to supply the hydraulic fluid to the propulsion device **160**, the swing motor **160**, and the working devices **170** such as boom cylinders, arm cylinders, bucket cylinders, and optional cylinders.

(16) The oil tank **850** supplies the hydraulic fluid to be discharged by the hydraulic pump **800**. The oil tank **850** also recovers the hydraulic fluid that has moved along the hydraulic line **610** from the hydraulic pump **800**.

- (17) A heating device **860** heats the hydraulic fluid stored in the oil tank **850**. For example, the heating device **860** may be an electric heating wire installed inside the oil tank **850**.
- (18) The main control valve (MCV) **500** is installed on the hydraulic line **610** and controls the supply of hydraulic fluid to one or more of the propulsion devices **160**, swing motor **180**, or working devices **170** that require hydraulic power. That is, the main control valve **500** distributes the hydraulic fluid discharged by the hydraulic pump **800** to various working devices **170**, swing motor **180**, and propulsion devices **160**, controlling the supply of hydraulic fluid.
- (19) In detail, the main control valve **500** includes a plurality of spools **510**. Each spool **510** controls the supply of hydraulic fluid to the propulsion devices **160**, swing motor **180**, and driving devices **170** such as boom cylinders, arm cylinders, bucket cylinders, and optional cylinders.
- (20) The main control valve **500** may also include spool caps (not shown) connected to each end of the control spools **510** to control the stroke of the control spools **510** based on the pilot signals received from the operating devices. For example, the spool caps may include electronic proportional pressure reducing valves (EPPRV), and depending on the degree of opening of the electronic proportional pressure reducing valve, the pressure of the pilot signal transmitted to the control spool **510** changes, causing the control spool **510** to move in both directions according to the pressure applied by the pilot signal.
- (21) The bypass cut valve **400** is installed downstream of the main control valve **500** on the hydraulic line **610** and is configured to open and close the hydraulic line **610**.
- (22) When the bypass cut valve **400** is in the closed position, the hydraulic fluid discharged from the hydraulic pump **800** is prevented from flowing through the hydraulic line **610** and returning to the oil tank **850**. In detail, the construction machine according to an embodiment includes a hydraulic pump **800** that is electronically controllable to prevent the return of hydraulic fluid when the bypass cut valve **400** is closed, maintaining a constant pressure level of 45 bar in the hydraulic circuit, and ensuring that the hydraulic fluid does not circulate. When the bypass cut valve **400** is switched to the open position, the hydraulic fluid discharged from the hydraulic pump **800** can return to the oil tank **850**.
- (23) That is, when the bypass cut valve **400** is in the closed position, even though the hydraulic pump **800** is in operation, the flow rate of hydraulic fluid moving along the hydraulic line **610** cannot increase.
- (24) The pilot cut-off lever **300** is installed inside the operator's seat of the construction machine, includes a lever and stand connected to each other, and controls whether the construction machine is able to operate. For example, the pilot cut-off lever **300** may be positioned in the first state (on state) to render the construction machine inoperable or in the second state (off state) to allow the construction machine to become operable. The first state may involve the lever being in a position perpendicular to the ground. The second state may involve the lever being in a position parallel to the ground. When the pilot cut-off lever **300** is in the first state, the pilot cut-off signal may be input, and when the pilot cut-off lever **300** is in the second state, the pilot cut-off signal may not be input.
- (25) The controller **700** controls various components of the construction machine, such as the electronic power unit **200**, main control valve **500**, and hydraulic pump **800**. The controller **700** may also include at least one of an electronic power unit control device **710** and a vehicle control device **720**.
- (26) In an embodiment of the present disclosure, the controller **700** may initiate a warm-up mode operation increasing the drive output of the electronic power unit **200** and opening the bypass cut valve **400**.
- (27) FIG. 2 is a block diagram of an electronic power unit according to an embodiment.
- (28) The electronic power unit **200** according to an embodiment includes a battery **210**, an inverter **230**, and an electric motor **250**.
- (29) The battery **210** is made of a rechargeable secondary battery. The secondary battery

encompasses nickel-metal hydride batteries, lithium-ion batteries, and lithium-polymer batteries. The secondary battery is composed of a positive electrode, a negative electrode, electrolyte, and a separator, and discharge occurs in the process of converting chemical energy into electrical energy. In temperatures below 0° C., the chemical reactions of the battery **210** slow down, resulting in degraded output characteristics and State of Charge (SOC) for the battery **210**.

(30) For example, in the case of a lithium-ion battery, electrical flow is generated through the movement of lithium ions between the positive and negative electrodes via the electrolyte, and at lower temperatures, the movement speed of lithium ions decreases. This results in increased internal resistance and reduced voltage in the battery **210**. Additionally, at temperatures below 0° C., the liquid electrolyte may freeze.

(31) The inverter **230** converts the direct current voltage from the battery **210** into the three-phase alternating current voltage required for the electric motor's operation and controls the torque of the electric motor **250**.

(32) The electric motor **250** is connected to the hydraulic pump **800** and transfers power to the hydraulic pump **800**. In detail, the electric motor **250** receives alternating current voltage from the inverter **230** to generate rotational power. The hydraulic pump **800** receives rotational power from the electric motor **250** to generate hydraulic pressure for the hydraulic fluid.

(33) The controller **700** may initiate the construction machine into a warm-up mode based on the temperature of the battery **210** or the hydraulic fluid received from the sensor **100**. The controller **700** may increase the output of the electronic power unit **200** in warm-up mode operation.

(34) FIG. 3 is a flowchart illustrating a control method of a controller.

(35) The controller **700** may operate the construction machine in warm-up mode or normal mode based on the temperature of the battery **210**, the temperature of the hydraulic fluid, and the pilot cut-off signal of the pilot cut-off lever. The warm-up mode is an operating mode of the construction machine for preventing the output of the construction machine from being degraded in low-temperature conditions. The controller **700** may increase the output of the electronic power unit **200** and control the bypass cut valve **400** to be open.

(36) Initially, when the construction machine is started, the controller **700** checks the temperature of the battery **210** at step S100. The controller **700** determines whether the temperature of the battery **210** is lower than the first threshold value Bth. The first threshold value Bth may be 0° C. Upon determining that the temperature of the battery **210** is lower than the first threshold value Bth, the controller **700** operates the electronic power unit **200** in battery **210** output limitation mode at step S300. Upon determining that the temperature of the battery **210** is higher than or equal to the first threshold value Bth, the controller **700** determines at step S200 whether the temperature of the hydraulic fluid is higher than a second threshold value Oth.

(37) The controller **700** may check the temperature of the hydraulic fluid. The controller **700** determines whether the temperature of the hydraulic fluid is higher than the second threshold value Oth. The second threshold value Oth may be 40° C. The viscosity of the hydraulic fluid, which varies with temperature, may cause issues with the responsiveness of the drive devices when the temperature is below 40° C. That is, even though the amount of hydraulic fluid discharged by the hydraulic pump is the same, the responsiveness of the driving devices may vary due to the viscosity differences in the hydraulic fluid based on its temperature. Upon determining that the temperature of the hydraulic fluid is lower than or equal to the second threshold value Oth, the controller **700** operates the electronic power unit **200** in the battery output limitation mode at step S200. Upon determining that the temperature of the battery **210** is higher than or equal to the first threshold value Bth, the controller **700** determines at step S200 whether the temperature of the hydraulic fluid is higher than a second threshold value Oth.

(38) Upon determining that the temperature of the hydraulic fluid is lower than or equal to the second threshold value Oth, the controller **700** may operate the heating device **860** additionally to run the warm-up mode. The heating device **860** may be an electric heating wire installed inside the

oil tank **850**, and the controller may operate the heating device **860** to heat and raise the temperature of the hydraulic fluid stored in the oil tank **850**.

(39) In the battery output limitation mode at step **S300**, the controller **700** may limit the output of the battery **210** to control the output of the electric motor **250** based on the temperature of the battery **210** and the temperature of the hydraulic fluid. In detail, the controller **700** may increase the discharge amount of the battery **210** to raise the temperature of the battery **210** in a state where the battery temperature is lower than the first threshold value B_{th} . The battery **210** has an internal resistance of tens of milliohms ($m\Omega$), and when discharging begins, self-heating occurs proportionally to the square of the discharge rate (C-rate) according to Joule's law. Therefore, it is possible to raise the temperature of the battery **210**. The controller **700** may adjust the output of the battery **210** based on the temperature conditions of the battery **210**. For example, the controller **700** may adjust the maximum discharge rate (Max. C-rate) of the battery **210** based on the temperature conditions of the battery **210**, as shown in Table 1. The upper limit of the output of the battery **210** at different temperatures may be compiled into a lookup table and included in the internal logic of the battery management system (BMS).

(40) TABLE-US-00001 TABLE 1 Maximum Battery discharge temperature rate ($^{\circ}$ C.) (C)
-30~-20 0.2 -20~-5 0.3 >-5 0.4

(41) The controller **700** may discharge the battery **210** by discharging the charge stored in the battery **210** to an external load. The controller **700** may increase the torque of the electric motor **250** supplied with electrical energy from the battery **210** or increase the rpm of the electric motor **250** to raise the discharge amount of the battery **210**. As the torque or rpm of the electric motor **250** increases, the hydraulic pressure generated by the hydraulic pump **800** receiving rotational power from the electric motor **250** increases. As the hydraulic pressure increases, the circulation of hydraulic fluid within the hydraulic lines becomes faster, and the temperature of the hydraulic fluid rises due to friction within the hydraulic lines. In this case, the controller **700** may control to increase the output of the electronic power unit **200** and open the bypass cut valve **400**.

(42) Next, the controller **700** determines at step **S400** whether the pilot cut-off signal of the pilot cut-off lever **300** is input. The pilot cut-off lever **300**, typically installed inside the operator's seat, controls the operability of the construction machine by initiating the input of the pilot cut-off signal in the first state (on state) and terminating the input of the pilot cut-off signal in the second state (off state). When the pilot cut-off lever **300** is in the first state (on state), the pilot cut-off signal is input to the controller **700**, rendering the construction machine to inoperable. When the pilot cut-off lever **300** is in the first state (on state), even though the hydraulic pressure generated by the hydraulic pump **800** increases, the propulsion devices **160**, swing motor **160**, and various working devices **170**, such as boom cylinders, arm cylinders, bucket cylinders, and optional cylinders, of the construction machine do not operate, ensuring safety.

(43) Upon determining at step **S400** that the pilot cut-off signal is input in the battery output limitation mode (**S300**), the controller **700** operates the construction machine in warm-up mode.

(44) Upon determining that the pilot cut-off signal is not input, the controller **700** determines that the pilot cut-off signal lever **300** is in the second state (off state) and controls the construction machine to generate an alarm at step **S500**. For example, the controller **700** may generate an alarm to the operator to operate the pilot cut-off lever **300** to the first state (on state) through the instrument panel of the construction machine. Additionally, the controller **700** may generate an alarm to the operator to operate the pilot cut-off lever **300** to the first state (on state) through a buzzer warning sound or voice.

(45) The controller **700** may check the temperature of the battery **210** in warm-up mode at step **S100**. Afterward, the controller **700** may switch the construction machine to normal mode or maintain normal operation based on the temperature of the battery **210**, the temperature of the hydraulic fluid, and the pilot cut-off signal.

(46) The above-described embodiments should be understood as illustrative and not limiting in all

respects. The scope of the present disclosure is determined by the appended claims rather than the specification above, and any modifications or variations derived from the meaning and scope of the claims and their equivalence should be interpreted as included the scope of the claims of the present disclosure.

Claims

1. A construction machine comprising: an electronic power unit comprising a battery and an electric motor; a hydraulic pump driven by the electronic power unit and discharging hydraulic fluid; a hydraulic line through which the hydraulic fluid discharged from the hydraulic pump moves; a main control valve installed on the hydraulic line and controlling the supply of the hydraulic fluid to at least one of propulsion devices or various working devices requiring hydraulic power; a bypass cut valve installed downstream of the main control valve on the hydraulic line and configured to open and close the hydraulic line; and a controller initiating a warm-up mode to increase output of the electronic power unit and open the bypass cut valve, wherein the controller: initiates the warm-up mode based on temperature of the battery being lower than a first threshold value; and during the warm-up mode, controls the battery by changing a discharge rate of the battery based on the temperature of the battery.
2. The construction machine of claim 1, wherein the controller initiates the warm-up mode based on temperature of the hydraulic fluid being lower than or equal to a second threshold value.
3. The construction machine of claim 2, wherein the controller, during the warm-up mode, additionally activates a heating device based on the temperature of the hydraulic fluid being lower than or equal to the second threshold value.
4. The construction machine of claim 1, wherein the controller initiates the warm-up mode based on a pilot cut-off signal being input from a pilot cut-off lever.
5. The construction machine of claim 4, wherein the controller controls the construction machine to generate an alarm based on the pilot cut-off signal being not input.
6. A method for controlling a construction machine comprising an electronic power unit comprising a battery and an electric motor, a hydraulic pump driven by the electronic power unit and discharging hydraulic fluid, a hydraulic line through which the hydraulic fluid discharged from the hydraulic pump moves, a main control valve installed on the hydraulic line and controlling the supply of the hydraulic fluid to at least one of propulsion devices or various working devices requiring hydraulic power, a bypass cut valve installed downstream of the main control valve on the hydraulic line and configured to open and close the hydraulic line, and an oil tank storing the hydraulic fluid to be supplied to the hydraulic pump and recovering the hydraulic fluid discharged by the hydraulic pump and moved along the hydraulic line, the method comprising initiating a warm-up mode to increase output of the electronic power unit and open the bypass cut valve, wherein the initiating of the warm-up mode comprises: initiating the warm-up mode based on temperature of the battery being lower than a first threshold value; and controlling the battery by changing a discharge rate of the battery based on the temperature of the battery.
7. The method of claim 6, wherein the initiating of the warm-up mode comprises initiating the warm-up mode based on temperature of the hydraulic fluid being lower than or equal to a second threshold value.
8. The method of claim 7, wherein the initiating of the warm-up mode comprises activating a heating device additionally based on the temperature of the hydraulic fluid being lower than or equal to the second threshold value.
9. The method of claim 6, wherein the initiating of the warm-up mode comprises initiating the warm-up mode based on a pilot cut-off signal being input from a pilot cut-off lever.

10. The method of claim 9, wherein the initiating of the warm-up mode comprises controlling the construction machine to generate an alarm based on the pilot cut-off signal being not input.
